

Handling sky localization maps

	Flat Maps	Multi-Order Maps (HEALPix Multi-Order, MOC)
Underlying Pixelization	HEALPix with fixed resolution (one Nside)	HEALPix with variable resolution (multi-order)
Storage Size	Always full sky → large files even if probability is localized	Only stores pixels that contain probability → 10–100× smaller
Resolution	Uniform resolution everywhere (wasteful in low-probability regions)	High resolution only where needed (e.g., 0.1° near source, degrees elsewhere)
Probability Representation	Fixed grid; downsampling loses information	Exact lossless representation at arbitrary resolution
Typical File Size	50–200 MB per event	50–500 kB per event

SCiMMA (Scalable Cyberinfrastructure for Multi-Messenger Astrophysics)

- A collaborative initiative supporting **real-time data sharing** across gravitational wave, gamma-ray, and neutrino observatories.
- Provides tools, services, and protocols to enable **rapid follow-up of transient astrophysical events**.
- Bridges distributed observatories, high-performance computing, and the wider astrophysics community.

<https://scimma.org/>

Hopskotch (Hop)

- **Pub-sub messaging system** developed by SCiMMA.
- **Built on Apache Kafka**, providing reliable and scalable real-time event streaming.
- Supports multiple topics (GCN alerts, GW candidate streams, telescope telemetry, etc.).
- Enables **programmatic access via Python or CLI** for subscribing, publishing, and monitoring streams.